Odd denominators in power-sum expansions
Stanley supplementary exercise 21

Statement: Richard P. Stanley
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Statement (Stanley, supplementary exercise 21)

Let $f \in \Lambda_{\mathbb{Z}} \cap \mathbb{Q}[p_1, p_3, p_5, \ldots]$, and write $f = \sum_\lambda a_\lambda p_\lambda$. Show that when the rational number a_λ is written in lowest terms, its denominator is odd.

Source correspondence. Stanley's solutions explicitly cite Darij Grinberg's answer to MathOverflow question 403153. The author's Theorem 1 below treats an arbitrary prime q; the original exercise uses its $q = 2$ case. All supporting results supplied in that answer are retained here.

Author answer (Darij Grinberg)

Here is a proof of the generalization suggested by Richard Stanley in the comments and even of a more general result (with "odd" replaced by "not divisible by a given prime q"). It is completely different from the argument I sketched in the comments, and is completely elementary (using no Macdonald polynomials). Unfortunately, it is also somewhat awkward and way too long (much of it devoted to a fight with notations).

For any commutative ring R, we let Λ_R be the ring of symmetric functions over R; this is a commutative R-algebra. Let Par be the set of all partitions. We shall use the standard notation p_λ for the power-sum symmetric function indexed by a partition λ.

Fix a prime q. Let $\mathbb{Z}_{(q)}$ denote the ring of all rational numbers that can be written in the form $\dfrac{a}{b}$ for two integers a and b such that b is coprime to q. These numbers are known as *q-integers*. Obviously, $\mathbb{Z}_{(q)}$ is a subring of $\mathbb{Q}$, so that $\Lambda_{\mathbb{Z}_{(q)}}$ is a subring of $\Lambda_{\mathbb{Q}}$.

Now, Richard Stanley's generalization (generalized a bit further) claims:

Theorem 1. We have

$$\Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right] = \mathbb{Z}_{(q)}\left[p_i \mid i \not\equiv 0 \mod q\right].$$

The proof requires some preparations, which include showing some results of independent interest.

First, we introduce a few more classical notations from symmetric function theory: For any partition λ and any $i \geq 1$, we let $m_i(\lambda)$ denote the multiplicity of i in λ (that is, the number of parts of λ that equal i). For any partition λ, we define the positive integer

$$z_\lambda := \prod_{i=1}^{\infty} \left((m_i(\lambda))! \cdot i^{m_i(\lambda)} \right).$$

Let V be the $\Lambda_{\mathbb{Z}}$-subalgebra of $\Lambda_{\mathbb{Q}}$ generated by the fractions $\dfrac{p_i}{i^k}$ for all positive integers i and all nonnegative integers k. In other words, let

$$V = \Lambda_{\mathbb{Z}}\left[\frac{p_i}{i^k} \mid i > 0 \text{ and } k \geq 0\right] \subseteq \Lambda_{\mathbb{Q}}.$$

Now, we claim the following:

Theorem 2. We have

$$z_\lambda^{-1} p_\lambda \in V$$

for each partition λ.

To prove this, we need a simple arithmetic lemma:

Lemma 3. Let c and d be two integers with $d \neq 0$. Then, there exist some integers a and b and some nonnegative integer i such that $c^i = ad + bc^{i+1}$.

Proof of Lemma 3. The ring $\mathbb{Z}/d\mathbb{Z}$ is finite (since $d \neq 0$). For any integer m, we let $\overline{m} \in \mathbb{Z}/d\mathbb{Z}$ denote the residue class of m in this ring. The infinitely many residue classes $\overline{c^0}, \overline{c^1}, \overline{c^2}, \ldots$ all belong to the finite ring $\mathbb{Z}/d\mathbb{Z}$, and thus cannot all be distinct (by the pigeonhole principle). In other words, there exist two nonnegative integers i and j satisfying $i < j$ and $\overline{c^i} = \overline{c^j}$. Consider these i and j. We have $\overline{c^i} = \overline{c^j}$; in other words, $c^i \equiv c^j \mod d$. Hence, $d \mid c^i - c^j$. In other words, $c^i - c^j = ad$ for some integer a. Consider this a. However, recall that $i < j$. Thus, $j = i + w$ for some positive integer w. Consider this w. The integer $w-1$ is nonnegative (since w is positive); thus, c^{w-1} is an integer. Hence, we can define an integer b by $b = c^{w-1}$. From $j = i + w = (w-1) + (i+1)$, we obtain $c^j = c^{(w-1)+(i+1)} = \underbrace{c^{w-1}}_{=b} c^{i+1} = bc^{i+1}$. Now, from $c^i - c^j = ad$, we obtain

$$c^i = ad + \underbrace{c^j}_{=bc^{i+1}} = ad + bc^{i+1}.$$

This proves Lemma 3. ■

Proof of Theorem 2. We shall use the notation $\ell(\lambda)$ for the *length* of a partition λ (that is, the number of all parts of λ). For instance, $\ell((5,2,2)) = 3$. We shall also use the notation $|\lambda|$ for the *size* of a partition λ (that is, the sum of all parts of λ). We shall prove Theorem 2 by strong induction on $|\lambda| + \ell(\lambda)$. Thus, we fix some $N \in \mathbb{N}$, and we assume (as the induction hypothesis) that Theorem 2 holds for all λ with $|\lambda| + \ell(\lambda) < N$. We now must prove Theorem 2 for all λ with $|\lambda| + \ell(\lambda) = N$.

So let λ be a partition satisfying $|\lambda| + \ell(\lambda) = N$. We must prove that $z_\lambda^{-1} p_\lambda \in V$.

If $\lambda = \varnothing$, then this is obvious (since $z_\lambda^{-1} p_\lambda = 1$ in this case). Thus, for the rest of this induction step, we WLOG assume that $\lambda \neq \varnothing$.

We are in one of the following three cases:

Case 1: All parts of λ are equal to 1.

Case 2: All parts of λ are equal, but not equal to 1.

Case 3: Not all parts of λ are equal.

Let us consider Case 1 first. In this case, all parts of λ are equal to 1. In other words, $\lambda = \underbrace{(1,1,\ldots,1)}_{v \text{ entries}}$ for some positive integer v (since $\lambda \neq \varnothing$). Consider this v. Thus, $p_\lambda = p_1^v$ and $|\lambda| = v$ and $\ell(\lambda) = v$. Let Par_v denote the set of all partitions of v. Thus, $\lambda \in \operatorname{Par}_v$. Now, let $h_v \in \Lambda_{\mathbb{Z}}$ denote the v-th complete homogeneous symmetric function. A well-known formula (e.g., (2.5.17) in Grinberg/Reiner, arXiv:1409.8356v7) yields

$$h_v = \sum_{\mu \in \operatorname{Par}_v} z_\mu^{-1} p_\mu = z_\lambda^{-1} p_\lambda + \sum_{\substack{\mu \in \operatorname{Par}_v; \\ \mu \neq \lambda}} z_\mu^{-1} p_\mu,$$

so that

$$z_\lambda^{-1} p_\lambda = h_v - \sum_{\substack{\mu \in \operatorname{Par}_v; \\ \mu \neq \lambda}} z_\mu^{-1} p_\mu. \tag{1}$$

However, the only partition of v that has length $\geq v$ is the partition $\underbrace{(1,1,\ldots,1)}_{v \text{ entries}} = \lambda$. Thus, if μ is a partition of v distinct from λ, then μ has length $< v$. In other words, if $\mu \in \operatorname{Par}_v$ satisfies $\mu \neq \lambda$, then $\ell(\mu) < v$. Hence, if $\mu \in \operatorname{Par}_v$ satisfies $\mu \neq \lambda$, then

$$\underbrace{|\mu|}_{=v=|\lambda|} + \underbrace{\ell(\mu)}_{<v=\ell(\lambda)} < |\lambda| + \ell(\lambda) = N$$

and therefore $z_\mu^{-1} p_\mu \in V$ (by our induction hypothesis, applied to μ instead of λ). Thus, (1) becomes

$$z_\lambda^{-1} p_\lambda = \underbrace{h_v}_{\in \Lambda_{\mathbb{Z}} \subseteq V} - \sum_{\substack{\mu \in \operatorname{Par}_v; \\ \mu \neq \lambda}} \underbrace{z_\mu^{-1} p_\mu}_{\in V} \in V - \sum_{\substack{\mu \in \operatorname{Par}_v; \\ \mu \neq \lambda}} V \subseteq V.$$

Hence, $z_\lambda^{-1} p_\lambda \in V$ has been proved in Case 1.

Let us next consider Case 2. In this case, all parts of λ are equal, but not equal to 1. In other words, $\lambda = \underbrace{(n,n,\ldots,n)}_{v \text{ entries}}$ for some positive integers $n \neq 1$ and v (since $\lambda \neq \varnothing$). Consider these n and v. Thus, $p_\lambda = p_n^v$ and $z_\lambda = v! \cdot n^v$ and $|\lambda| = nv$ and $\ell(\lambda) = v$. From $n \neq 1$, we obtain $n > 1$ (since n is a positive integer). Thus, $nv > v$ (since $v > 0$). In other words, $v < nv$.

From $p_\lambda = p_n^v$ and $z_\lambda = v! \cdot n^v$, we obtain

$$z_\lambda^{-1} p_\lambda = (v! \cdot n^v)^{-1} p_n^v = \frac{p_n^v}{v! \cdot n^v}. \tag{2}$$

We must prove that $z_\lambda^{-1} p_\lambda \in V$. In other words, we must prove that $\dfrac{p_n^v}{v! \cdot n^v} \in V$ (since $z_\lambda^{-1} p_\lambda = \dfrac{p_n^v}{v! \cdot n^v}$).

We shall now use Lemma 3 to decompose $\dfrac{p_n^v}{v! \cdot n^v}$ into a sum of partial fractions – one with a denominator of $v!$ and another with a power of n in the denominator. We will then prove that both of these fractions belong to V.

Indeed, Lemma 3 (applied to $c = n^v$ and $d = v!$) yields that there exist some integers a and b and some nonnegative integer i such that $(n^v)^i = a \cdot v! + b\,(n^v)^{i+1}$. Consider these a, b and i. Multiplying both sides of the equality $(n^v)^i = a \cdot v! + b\,(n^v)^{i+1}$ by $\dfrac{p_n^v}{(n^v)^{i+1} \cdot v!}$, we obtain

$$\frac{p_n^v}{v! \cdot n^v} = \left(a \cdot v! + b\,(n^v)^{i+1}\right) \cdot \frac{p_n^v}{(n^v)^{i+1} \cdot v!} \tag{1}$$

$$= a \cdot \frac{p_n^v}{(n^v)^{i+1}} + b \cdot \frac{p_n^v}{v!}. \tag{3}$$

Thus, in order to prove that $\dfrac{p_n^v}{v! \cdot n^v} \in V$, it suffices to show that $\dfrac{p_n^v}{(n^v)^{i+1}} \in V$ and $\dfrac{p_n^v}{v!} \in V$ (because a and b are integers, and V is a ring).

The first of these two claims is easy: We have $\dfrac{p_n}{n^{i+1}} \in V$ (since $\dfrac{p_n}{n^{i+1}}$ is one of the designated generators of the $\Lambda_{\mathbb{Z}}$-algebra V). Hence, $\left(\dfrac{p_n}{n^{i+1}}\right)^v \in V$ (since V is a ring). In other words, $\dfrac{p_n^v}{(n^v)^{i+1}} \in V$ (since $\left(\dfrac{p_n}{n^{i+1}}\right)^v = \dfrac{p_n^v}{(n^v)^{i+1}}$).

It remains to prove that $\dfrac{p_n^v}{v!} \in V$. To do this, we will need the *Frobenius endomorphism* $\mathbf{f}_n$. It is defined as follows: For any commutative ring R, we let

$$\mathbf{f}_n : \Lambda_R \to \Lambda_R$$

be the R-algebra homomorphism that sends each symmetric function f to $f(x_1^n, x_2^n, x_3^n, \ldots)$ (where we regard f as a symmetric formal power series in countably many indeterminates $x_1, x_2, x_3, \ldots$). This homomorphism $\mathbf{f}_n$ is called the n-th *Frobenius endomorphism* and is functorial in R (that is, it commutes with the morphisms $\Lambda_R \to \Lambda_S$ induced by ring homomorphisms $R \to S$). If you like to think in terms of plethysm, $\mathbf{f}_n$ can be described as sending each $f \in \Lambda_R$ to the plethysm $f[p_n]$.

The functoriality of $\mathbf{f}_n$ in R entails that the $\mathbf{f}_n$ defined for $R = \mathbb{Z}$ is a restriction of the $\mathbf{f}_n$ defined for $R = \mathbb{Q}$. Thus, we can safely denote both of these maps by $\mathbf{f}_n$ without risking confusion. They both are ring homomorphisms (since they are R-algebra homomorphisms for appropriate R). Of course, $\mathbf{f}_n(\Lambda_{\mathbb{Z}}) \subseteq \Lambda_{\mathbb{Z}}$ (since the $\mathbf{f}_n$ defined for $R = \mathbb{Z}$ is a restriction of the $\mathbf{f}_n$ defined for $R = \mathbb{Q}$).

It is easy to see that $\mathbf{f}_n(p_i) = p_{in}$ for each $i > 0$. Hence, for each positive integer i and each nonnegative integer k, we have

$$\mathbf{f}_n\left(\frac{p_i}{i^k}\right) = \frac{p_{in}}{i^k} = n^k \cdot \frac{p_{in}}{(in)^k} \in V$$

(since $\dfrac{p_{in}}{(in)^k}$ is one of the designated generators of the $\Lambda_{\mathbb{Z}}$-algebra V). Therefore,

$$\Lambda_{\mathbb{Z}}\left[\mathbf{f}_n\left(\frac{p_i}{i^k}\right) \;\mid\; i > 0 \text{ and } k \geq 0\right] \subseteq V$$

(since V is a $\Lambda_{\mathbb{Z}}$-algebra).

Now, from $V = \Lambda_{\mathbb{Z}}\left[\dfrac{p_i}{i^k} \;\mid\; i > 0 \text{ and } k \geq 0\right]$, we obtain

$$\begin{aligned}
\mathbf{f}_n(V) &= \mathbf{f}_n\left(\Lambda_{\mathbb{Z}}\left[\frac{p_i}{i^k} \;\mid\; i > 0 \text{ and } k \geq 0\right]\right) \\
&= \underbrace{\left(\mathbf{f}_n(\Lambda_{\mathbb{Z}})\right)}_{\subseteq \Lambda_{\mathbb{Z}}}\left[\mathbf{f}_n\left(\frac{p_i}{i^k}\right) \;\mid\; i > 0 \text{ and } k \geq 0\right] \\
&\qquad \text{(since } \mathbf{f}_n \text{ is a ring homomorphism)} \\
&\subseteq \Lambda_{\mathbb{Z}}\left[\mathbf{f}_n\left(\frac{p_i}{i^k}\right) \;\mid\; i > 0 \text{ and } k \geq 0\right] \subseteq V.
\end{aligned}$$

Let μ be the partition $\underbrace{(1,1,\ldots,1)}_{v \text{ entries}}$. Then, $p_\mu = p_1^v$ and $z_\mu = v! \cdot \underbrace{1^v}_{=1} = v!$ and $|\mu| = v$ and $\ell(\mu) = v$. Hence,

$$\begin{aligned}\underbrace{|\mu|}_{=v} + \underbrace{\ell(\mu)}_{=v} &= \underbrace{v}_{<nv} + v \\ &< \underbrace{nv}_{=|\lambda|} + \underbrace{v}_{=\ell(\lambda)} = |\lambda| + \ell(\lambda) = N.\end{aligned}$$

Hence, $z_\mu^{-1} p_\mu \in V$ (by our induction hypothesis, applied to μ instead of λ). In view of $z_\mu = v!$ and $p_\mu = p_1^v$, this rewrites as $v!^{-1} \cdot p_1^v \in V$. Hence,

$$\mathbf{f}_n\left(v!^{-1} \cdot p_1^v\right) \in \mathbf{f}_n(V) \subseteq V. \tag{4}$$

However, since $\mathbf{f}_n$ is a $\mathbb{Q}$-algebra homomorphism, we have

$$\mathbf{f}_n\left(v!^{-1} \cdot p_1^v\right) = v!^{-1} \cdot \left(\mathbf{f}_n(p_1)\right)^v = \frac{\left(\mathbf{f}_n(p_1)\right)^v}{v!} = \frac{p_n^v}{v!}$$

(since $\mathbf{f}_n(p_1) = p_n$). Thus, (4) rewrites as $\frac{p_n^v}{v!} \in V$. Hence, (3) becomes

$$\frac{p_n^v}{v! \cdot n^v} = a \cdot \underbrace{\frac{p_n^v}{\left(n^v\right)^{i+1}}}_{\in V} + b \cdot \underbrace{\frac{p_n^v}{v!}}_{\in V} \in V$$

(since V is a ring and since a and b are integers). In view of (2), this rewrites as $z_\lambda^{-1} p_\lambda \in V$. Hence, $z_\lambda^{-1} p_\lambda \in V$ has been proved in Case 2.

Let us finally consider Case 3. In this case, not all parts of λ are equal.

We need another notation: If $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_n)$ are two partitions, then $\alpha \sqcup \beta$ shall denote the partition obtained by sorting the tuple $(\alpha_1, \alpha_2, \ldots, \alpha_m, \beta_1, \beta_2, \ldots, \beta_n)$ in weakly decreasing order. For instance, $(3,2,2) \sqcup (5,3,2) = (5,3,3,2,2,2)$.

It is easy to see that if α and β are two partitions that have no part in common, then

$$z_{\alpha \sqcup \beta} = z_\alpha z_\beta. \tag{5}$$

Moreover, if α and β are any two partitions, then

$$p_{\alpha \sqcup \beta} = p_\alpha p_\beta. \tag{6}$$

Now, recall that not all parts of λ are equal. Hence, we can write λ in the form $\lambda = \alpha \sqcup \beta$ where α and β are two nonempty partitions that have no part in common. (Indeed, we can define α and β by choosing an arbitrary part i of λ, then letting α be the partition consisting of all parts of λ equal to i, while β is the partition consisting of all remaining parts of λ.) Consider these α and β. It is easy to see that $|\alpha| < |\alpha \sqcup \beta|$ (since β is nonempty) and $\ell(\alpha) < \ell(\alpha \sqcup \beta)$ (for the same reason). Since $\alpha \sqcup \beta = \lambda$, these two inequalities rewrite as $|\alpha| < |\lambda|$ and $\ell(\alpha) < \ell(\lambda)$. Adding these two inequalities together, we obtain

$$|\alpha| + \ell(\alpha) < |\lambda| + \ell(\lambda) = N.$$

Hence, $z_\alpha^{-1} p_\alpha \in V$ (by our induction hypothesis, applied to α instead of λ). Similarly, $z_\beta^{-1} p_\beta \in V$. However, from (5) and (6), we obtain

$$z_{\alpha \sqcup \beta}^{-1} p_{\alpha \sqcup \beta} = \left(z_\alpha z_\beta\right)^{-1} p_\alpha p_\beta = \underbrace{z_\alpha^{-1} p_\alpha}_{\in V} \cdot \underbrace{z_\beta^{-1} p_\beta}_{\in V} \in V$$

(since V is a ring). In view of $\alpha \sqcup \beta = \lambda$, this rewrites as $z_\lambda^{-1} p_\lambda \in V$. Hence, $z_\lambda^{-1} p_\lambda \in V$ has been proved in Case 3.

We have now proved $z_\lambda^{-1} p_\lambda \in V$ in all three cases 1, 2 and 3. Thus, the induction step is complete.

Thus, Theorem 2 is proved by induction. ■

We are not quite ready to prove Theorem 1 yet. We first need some more notations.

We let QPar be the set of all partitions that have no part divisible by q. (If $q = 2$, these are precisely the partitions into odd parts.)

We let J be the ideal of the ring $\Lambda_\mathbb{Q}$ generated by the p_i with $i \equiv 0 \mod q$. In other words, $J = \sum_{i=1}^{\infty} p_{iq} \Lambda_\mathbb{Q}$. Recall that the family $(p_\lambda)_{\lambda \in \mathrm{Par}}$ is a basis of the $\mathbb{Q}$-vector space $\Lambda_\mathbb{Q}$. Thus, the $\mathbb{Q}$-vector subspace J of $\Lambda_\mathbb{Q}$ has basis $(p_\lambda)_{\lambda \in \mathrm{Par} \setminus \mathrm{QPar}}$ (because multiplying any p_μ by a p_{iq} yields a p_λ with $\lambda \in \mathrm{Par} \setminus \mathrm{QPar}$, and conversely, any p_λ with $\lambda \in \mathrm{Par} \setminus \mathrm{QPar}$ can be obtained in such a way).

A well-known fact (or easy exercise) in abstract algebra says the following:

Lemma 4. Let B be a subring of a ring A. Let I be a (two-sided) ideal of A. Then, $B+I$ is a subring of A.

Applying Lemma 4 to $A = \Lambda_{\mathbb{Q}}$, $B = \Lambda_{\mathbb{Z}_{(q)}}$ and $I = J$, we conclude that $\Lambda_{\mathbb{Z}_{(q)}} + J$ is a subring of $\Lambda_{\mathbb{Q}}$. We denote this subring $\Lambda_{\mathbb{Z}_{(q)}} + J$ by W. We note that W is furthermore a $\mathbb{Z}_{(q)}$-subalgebra of $\Lambda_{\mathbb{Q}}$ (since W is a subring of $\Lambda_{\mathbb{Q}}$ and is preserved under scaling by $\mathbb{Z}_{(q)}$).

Next, we observe:

Proposition 5. We have $V \subseteq W$.

Proof of Proposition 5. We have

$$\Lambda_{\mathbb{Z}} \subseteq \Lambda_{\mathbb{Z}_{(q)}} \subseteq \Lambda_{\mathbb{Z}_{(q)}} + J = W. \tag{2}$$

Now, W is a commutative ring (since it is a subring of $\Lambda_{\mathbb{Q}}$) and contains $\Lambda_{\mathbb{Z}}$ as a subring (since $\Lambda_{\mathbb{Z}} \subseteq W$). Hence, W is a $\Lambda_{\mathbb{Z}}$-algebra. Thus, in order to prove that $V \subseteq W$, it suffices to show that $\dfrac{p_i}{i^k} \in W$ for each positive integer i and each nonnegative integer k (by the definition of V).

So let us show this. Fix a positive integer i and a nonnegative integer k. We must prove that $\dfrac{p_i}{i^k} \in W$. If $i \equiv 0 \bmod q$, then this follows from the obvious fact that $\dfrac{p_i}{i^k} \in J \subseteq \Lambda_{\mathbb{Z}_{(q)}} + J = W$. Thus, we WLOG assume that $i \not\equiv 0 \bmod q$. Hence, i is coprime to q. Hence, $\dfrac{1}{i} \in \mathbb{Z}_{(q)}$, so that $\dfrac{1}{i^k} \in \mathbb{Z}_{(q)} \subseteq W$. Now, $\dfrac{p_i}{i^k} = \underbrace{\left(\dfrac{1}{i}\right)^k}_{\in W} \underbrace{p_i}_{\in \Lambda_{\mathbb{Z}} \subseteq W} \in W$ (since W is a ring). This completes our proof of Proposition 5. ■

At last, we can prove Theorem 1.

Proof of Theorem 1. It is clear that $\mathbb{Z}_{(q)}\left[p_i \mid i \not\equiv 0 \mod q\right] \subseteq \Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right]$. Hence, it suffices to prove the reverse inclusion, i.e., to prove that

$$\Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right] \subseteq \mathbb{Z}_{(q)}\left[p_i \mid i \not\equiv 0 \mod q\right].$$

Thus, we fix an arbitrary $f \in \Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right]$. We must prove that $f \in \mathbb{Z}_{(q)}\left[p_i \mid i \not\equiv 0 \mod q\right]$.

We have $f \in \Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right] \subseteq \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right]$. Hence, f is a $\mathbb{Q}$-linear combination of the family $(p_\lambda)_{\lambda \in \mathrm{QPar}}$ (since this family $(p_\lambda)_{\lambda \in \mathrm{QPar}}$ is a basis of the $\mathbb{Q}$-vector space $\mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right]$). In other words, we can write f in the form

$$f = \sum_{\lambda \in \mathrm{QPar}} c_\lambda p_\lambda \tag{7}$$

for some $c_\lambda \in \mathbb{Q}$. Consider these c_λ. We shall prove that they all belong to $\mathbb{Z}_{(q)}$.

We let $\langle \cdot, \cdot \rangle$ denote the Hall inner product on $\Lambda_{\mathbb{Q}}$. This is a $\mathbb{Q}$-bilinear form sending $\Lambda_{\mathbb{Q}} \times \Lambda_{\mathbb{Q}}$ to $\mathbb{Q}$ and sending $\Lambda_{\mathbb{Z}_{(q)}} \times \Lambda_{\mathbb{Z}_{(q)}}$ to $\mathbb{Z}_{(q)}$ (since its restriction to $\Lambda_{\mathbb{Z}_{(q)}} \times \Lambda_{\mathbb{Z}_{(q)}}$ is the Hall inner product on $\Lambda_{\mathbb{Z}_{(q)}}$).

It is well-known (see, e.g., Corollary 2.5.17(b) in Grinberg/Reiner, arXiv:1409.8356v7) that the families $(p_\lambda)_{\lambda \in \mathrm{Par}}$ and $\left(z_\lambda^{-1} p_\lambda\right)_{\lambda \in \mathrm{Par}}$ are dual bases of $\Lambda_{\mathbb{Q}}$ with respect to the Hall inner product. Hence,

$$\left\langle p_\lambda, z_\mu^{-1} p_\mu \right\rangle = \delta_{\lambda,\mu} \tag{8}$$

for any $\lambda \in \mathrm{Par}$ and $\mu \in \mathrm{Par}$ (where the $\delta_{\lambda,\mu}$ is a Kronecker delta). In other words,

$$\left\langle p_\lambda, p_\mu \right\rangle = z_\mu \delta_{\lambda,\mu} \tag{9}$$

for any $\lambda \in \mathrm{Par}$ and $\mu \in \mathrm{Par}$.

Now, fix a partition $\mu \in \mathrm{QPar}$. Then, Theorem 2 (applied to $\lambda = \mu$) yields $z_\mu^{-1} p_\mu \in V \subseteq W$ (by Proposition 5). Hence, $z_\mu^{-1} p_\mu \in W = \Lambda_{\mathbb{Z}_{(q)}} + J$. In other words, we can write $z_\mu^{-1} p_\mu$ in the form $z_\mu^{-1} p_\mu = w_1 + w_2$ for some $w_1 \in \Lambda_{\mathbb{Z}_{(q)}}$ and some $w_2 \in J$. Consider these w_1 and w_2.

Now, it is easy to see that $\langle f, w_1 \rangle \in \mathbb{Z}_{(q)}$. [*Proof:* We have $f \in \Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q}\left[p_i \mid i \not\equiv 0 \mod q\right] \subseteq \Lambda_{\mathbb{Z}_{(q)}}$ and $w_1 \in \Lambda_{\mathbb{Z}_{(q)}}$. Thus, $(f, w_1) \in \Lambda_{\mathbb{Z}_{(q)}} \times \Lambda_{\mathbb{Z}_{(q)}}$. Hence, $\langle f, w_1 \rangle \in \mathbb{Z}_{(q)}$, because the Hall inner product $\langle \cdot, \cdot \rangle$ sends $\Lambda_{\mathbb{Z}_{(q)}} \times \Lambda_{\mathbb{Z}_{(q)}}$ to $\mathbb{Z}_{(q)}$.]

Furthermore, it is easy to see that $\langle f, w_2 \rangle = 0$. [*Proof:* We have $w_2 \in J$; thus, we can write w_2 as a $\mathbb{Q}$-linear combination of the family $(p_\lambda)_{\lambda \in \mathrm{Par} \setminus \mathrm{QPar}}$ (since this family is a basis of the $\mathbb{Q}$-vector space J). In other words, we can write w_2 in the form

$$w_2 = \sum_{\beta \in \mathrm{Par} \setminus \mathrm{QPar}} d_\beta p_\beta \tag{10}$$

for some coefficients $d_\beta \in \mathbb{Q}$. Consider these d_β. From (7) and (10), we obtain

$$\langle f, w_2 \rangle = \left\langle \sum_{\lambda \in \mathrm{QPar}} c_\lambda p_\lambda, \sum_{\beta \in \mathrm{Par} \setminus \mathrm{QPar}} d_\beta p_\beta \right\rangle$$

$$= \sum_{\lambda \in \mathrm{QPar}} \sum_{\beta \in \mathrm{Par} \setminus \mathrm{QPar}} c_\lambda d_\beta \underbrace{\langle p_\lambda, p_\beta \rangle}_{\substack{= z_\beta \delta_{\lambda,\beta} \\ \text{(by (9))}}}$$

$$= \sum_{\lambda \in \mathrm{QPar}} \sum_{\beta \in \mathrm{Par} \setminus \mathrm{QPar}} c_\lambda d_\beta z_\beta \underbrace{\delta_{\lambda,\beta}}_{\substack{=0 \\ \text{(since } \lambda \neq \beta \\ \text{(because } \lambda \in \mathrm{QPar} \\ \text{whereas } \beta \in \mathrm{Par} \setminus \mathrm{QPar}\text{))}}}$$

$$= \sum_{\lambda \in \mathrm{QPar}} \sum_{\beta \in \mathrm{Par} \setminus \mathrm{QPar}} c_\lambda d_\beta z_\beta 0 = 0,$$

qed.]

From $z_\mu^{-1} p_\mu = w_1 + w_2$, we obtain

$$\left\langle f, z_\mu^{-1} p_\mu \right\rangle = \langle f, w_1 + w_2 \rangle = \underbrace{\langle f, w_1 \rangle}_{\in \mathbb{Z}_{(q)}} + \underbrace{\langle f, w_2 \rangle}_{=0} \in \mathbb{Z}_{(q)}.$$

On the other hand, from (7), we obtain

$$\left\langle f, z_\mu^{-1} p_\mu \right\rangle = \left\langle \sum_{\lambda \in \mathrm{QPar}} c_\lambda p_\lambda, z_\mu^{-1} p_\mu \right\rangle = \sum_{\lambda \in \mathrm{QPar}} c_\lambda \underbrace{\left\langle p_\lambda, z_\mu^{-1} p_\mu \right\rangle}_{\substack{= \delta_{\lambda,\mu} \\ \text{(by (8))}}} = \sum_{\lambda \in \mathrm{QPar}} c_\lambda \delta_{\lambda,\mu}$$

$$= c_\mu.$$

Hence,

$$c_\mu = \left\langle f, z_\mu^{-1} p_\mu \right\rangle \in \mathbb{Z}_{(q)}.$$

Forget that we fixed μ. We thus have shown that $c_\mu \in \mathbb{Z}_{(q)}$ for each $\mu \in \mathrm{QPar}$. In other words, $c_\lambda \in \mathbb{Z}_{(q)}$ for each $\lambda \in \mathrm{QPar}$. Hence, (7) becomes

$$f = \sum_{\lambda \in \mathrm{QPar}} \underbrace{c_\lambda}_{\in \mathbb{Z}_{(q)}} p_\lambda \in \sum_{\lambda \in \mathrm{QPar}} \mathbb{Z}_{(q)} p_\lambda = \mathbb{Z}_{(q)} \left[p_i \mid i \not\equiv 0 \mod q \right]$$

(by the definition of QPar).

Forget that we fixed f. We thus have shown that $f \in \mathbb{Z}_{(q)} \left[p_i \mid i \not\equiv 0 \mod q \right]$ for each $f \in \Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q} \left[p_i \mid i \not\equiv 0 \mod q \right]$. In other words,

$$\Lambda_{\mathbb{Z}_{(q)}} \cap \mathbb{Q} \left[p_i \mid i \not\equiv 0 \mod q \right] \subseteq \mathbb{Z}_{(q)} \left[p_i \mid i \not\equiv 0 \mod q \right].$$

As explained, this completes the proof of Theorem 1. ■

Sources

Statement: R. P. Stanley, *Supplementary Exercises for Chapter 7*, version 4 June 2023, exercise 21, physical page 4; `https://math.mit.edu/~rstan/ec/ch7supp.pdf`. Official solution locator: version 24 March 2023, physical page 6; `https://math.mit.edu/~rstan/ec/ch7suppsol.pdf`.

Proof: Darij Grinberg, MathOverflow answer 403672 to question 403153, created 11 September 2021, edited 30 September 2021. `https://mathoverflow.net/a/403672`. The API records the answer license as CC BY-SA 4.0: `https://creativecommons.org/licenses/by-sa/4.0/`. The original HTML/MathJax answer is retained alongside this editable typeset transcription. The statement and answer have separate authors and source records.